

Bioinformatics Lab1

BioInf 2025/2026 1st Semester

André Lamúrias

a.lamurias@fct.unl.pt

Rosalind

- Platform to practice bioinformatics problems
- Problems are curated to appeal to people with various backgrounds
- For each problem, a description and example are given, as well as a new file to test your code
- You write the code to solve the problem (in any language you want) and upload the solution
- The website verifies if your solution is correct and allows multiple attempts

Rosalind

- For some practical classes we will use Rosalind: there will be a set of problems for you to solve related to the current topic
- These are not mandatory but will help you understand the topics of the lectures
- It will also help you with the practical projects and tests
- Problems of the week are set with a deadline but you can also finish them afterwards (delete “?class=1150” from URL)
- You can explore the other problems by yourself too

Rosalind

1. Create an account
2. Enroll in class
3. Solve problems that are still within the deadline
 1. Open problem
 2. Read description
 3. Write your solution, test with sample dataset
 4. Download test dataset file, then you have 5 minutes to upload the solution
 5. Run your code on the file and upload your output
 6. Check the result
4. If you want to solve a problem after its deadline, you can still find it
<https://rosalind.info/problems/<problem id>>

Resources

- Advice: disable copilot/autocomplete
 - Simple problems can be solved easily just by pressing “tab” but then you won’t be able to solve harder problems
- You can still use ChatGPT, as long as you don’t copy to problem description
 - I.e. you can use it the same way you can google or ask someone
- If you are stuck, ask your teacher before pasting the problem into ChatGPT or searching on the web

Resources

- You can use any language even though I'll show python examples in the lecture slides
- The algorithms don't require huge computational resources
- You can also use notebook, colab, etc
- Download the PDF of the exercises from moodle so that you can still solve them if you are offline (will be updated)

Rosalind tips

- Input files are named “rosalind_<id>.txt” so you can write code to read from this file even before testing
 - You can create this file and copy paste the example input
- You can also write the output to a file and upload this; sometimes the output is too big to be copy pasted
- Some problems have additional debug datasets
- Check the lecture slides that refer to a certain problem
- You can use the questions forum and/or share you solution to the solutions forum of each problem (only accessible to those enrolled)

Today: Lab1

- “Warm-up” exercises (Sept. 24, 2025 deadline)
- [DNA](#): Counting DNA Nucleotides
- [BA1A](#): Count k-mer pattern in string with overlaps
- [BA1B](#): Determine most frequent k-mers in string
- BA1C: Reverse Complement
- BA1D: Find all occurrences specific pattern in string
- BA1E: Find clumps of repeated k-mers
- BA1F: Find position that minimizes skew
- BA1G: Hamming distance (number of mismatches)
- BA1H: Find all approximate matches
- BA1I: Find most frequent k-mers with mismatches

Extra exercises

- BA1J: Find most frequent k-mers with mismatches and reverse complement
- BA1K: Generate the Frequency Array of a String